

Section 90 — Master Polynomial Algebra Mixed Contest

Contest contract

Six new problems combine the polynomial and transform tools of Sections 88–89.

Problem	Main decision
A	division as multiplication by a formal inverse
B	logarithm extracts connected components
C	product-tree evaluation with repeated queries
D	interpolation in arbitrary input order
E	OR zeta transform and Möbius inversion
F	rank-refined subset convolution

Do not choose a transform merely because masks or polynomials appear. State the binary operation that combines indices, then choose the transform whose characters diagonalize that operation.

Recognition workflow

1. Write the requested coefficient as an explicit sum over input indices.
2. Identify how those indices combine: addition, XOR, OR, AND, or disjoint union.
3. Check algebraic preconditions such as an invertible constant coefficient or distinct interpolation points.
4. Only then select multiplication, an FPS operation, a product tree, or a mask transform.

For connected-series problems, the decisive clue is an exponential assembly: an arbitrary object is a set of connected components. Taking a formal logarithm reverses that assembly and isolates the connected contribution. This is not a generic instruction to apply $\log$ whenever the word “connected” appears; the exact generating-function identity must be established first.

For OR convolution, define the subset-zeta transform

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Fhat[S] = sum of F[T] over all T contained in S.
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The product $\mathbf{Fhat}[S] * \mathbf{Ghat}[S]$ counts all pairs with both masks contained in S , equivalently all pairs whose OR is contained in S . Subset Möbius inversion then isolates pairs whose OR is exactly each mask. The analogous unranked argument is insufficient for disjoint union, because two submasks may overlap; popcount ranks expose and remove that ambiguity.

Oracle map

- Use naive coefficient recurrence for inverse and small FPS operations.
- Use Horner independently at every evaluation point.
- Enumerate the Lagrange basis directly for interpolation.
- Use a quadratic pair loop for XOR/OR convolution.
- Enumerate every ordered pair of submasks for disjoint covers.

Also test the smallest legal precision, zero-heavy polynomials, repeated evaluation points, shuffled interpolation input, the empty mask, and maximum mask dimensions with sparse nonzero entries.

Exercises

A–F form the complete mixed contest; all are new and self-contained.